Architecture Clustering. From the experiments by Boix-Adsera et al. [17], we consider the results depicted in Figure 16, where we consider the pretrained and untrained models as different variants and rank measures by their average standard deviation ratio.

Multilingual. From the multilingual benchmark by Rahamim and Belinkov [116], we considered the results depicted in Tables 2, 4, 5, where, in every test variant and for each probing layer, ContraSim had higher accuracy than CKA.

Image Caption. From the image caption benchmark by Rahamim and Belinkov [116], we considered the results depicted in Figure 5 and Table 3 and 6, where, again, ContraSim had higher accuracy than CKA in all test variants.

Shortcut Affinity. We considered and aggregated all results for test 4 (shortcut affinity) of `ReSi` [70] as presented in Appendix B, where we ranked all measures based on the reported AUPRC scores.

Augmentation. We considered and aggregated all results for test 5 (augmentation) of the `ReSi` benchmark [70] as presented in Appendix B, where we ranked all measures based on the reported AUPRC scores.

Label Randomization. We considered and aggregated all results for test 3 (label randomization) of the `ReSi` benchmark [70] as presented in Appendix B, where we ranked all measures based on the reported AUPRC scores.

Layer Monotonicity. We considered and aggregated all results for test 6 (layer monotonicity) of the `ReSi` benchmark [70] as presented in Appendix B, where we ranked all measures based on the reported Spearman correlation.

E Neural Network Similarity Beyond This Survey

In this survey, we reviewed representational similarity measures that can compare representations from two different models that use the same inputs, and functional similarity measures that compare models in (multi-class) classification contexts. Beyond the scope of this survey, there are other views on neural network similarity and application contexts, which we briefly discuss here.

Functional Similarity for Non-Classification Tasks. Although we focus on functional similarity with respect to classification, many of the functional similarity measures can be used for or directly transferred to other downstream tasks. In particular, if a suitable performance measure is given, performance-based measures can be used in any other context. This is also the case for gradient-based and stitching measures if white-box access to the models is given, and, in case of gradient-based measures, adversarial examples can be constructed for the given context. Soft and hard prediction-based measures, conversely, are limited to tasks where outputs are assigned discrete labels. For regression, one could consider binning outputs to obtain discrete labels. Further, there are specialized measures of agreement of continuous outputs [143, 46]. Finally, if output is structured, e.g., text or image generation, functional similarity becomes more difficult as outputs do not share universally identical semantics as in classification tasks. For example, generated images may have differences that are not perceivable to the human eye, and thus could be considered equivalent. This equivalence could lead to considering the invariances of functional similarity measures. The evaluation of these kinds of models, including comparison of outputs to a human reference, was studied in prior surveys [18, 22].

Alternative Notions of Neural Network Similarity. Aside from representational and functional similarity measures, there are several other notions of similarity that have been used to compare neural networks. Some of these approaches are applicable for specific types of neural networks. For instance, *visualizations* have emerged as a popular tool to analyze CNN similarity, although not limited to them. Approaches include the visualization of decision regions [220], neuron activations [169], or reconstructed images [203]. Chen et al. [174] proposed a method to compare *weights* of convolutional layers. For language models, *probing* [170] has become a popular approach. The idea behind probing is to compare the extent to which representations of models trained for a specific task such as sentiment analysis can also be used to predict related concepts such as part of speech.

There are also more universal approaches. For instance, Wang et al. [151] and Guth and Ménard [184] proposed methods to compare the *weight matrices* of neural networks. Further, one could also consider the impact of inputs, as done by Shah et al. [215]. They utilize the concept of *datamodels* [190], which aim to explain predictions in terms of which data samples were used in training. Using that approach, they measure similarity in neural networks by comparing the influence that data points have on individual predictions. Salle et al. [214] considered the extent to which differences in meta-features, such as part of speech or tense for text models, predict differences in instance representations, and compared different models based on the importance of such features for the prediction. Finally, measures that compare representations which are derived from different sets of inputs, but mapped into the same vector space, e.g., by coming from the same model, were proposed in the context of evaluating generative adversarial networks [200, 168] and metric learning [199, 192].

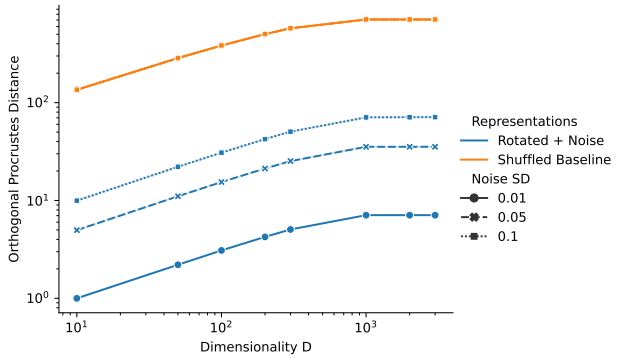

Figure 5: Mean Orthogonal Procrustes scores between two matrices over increasing dimensionality with varying noise level. The matrices have $N = 1000$ rows. Shuffled Baseline refers to the score between two effectively unrelated matrices, a row-wise shuffled copy of the representation matrix and the original, similar to Kriegeskorte et al. [74]. The baseline is unrelated to the noise level. Scores increase until the number of dimensions matches the number of inputs ($N = D$), then stays flat. While $N > D$, the relation between the similarity score and the dimensionality follows a power law, as shown by the linear relation in the log-log plot. The standard deviation is too small to be visible. The same trend can be observed with other N (not shown).

Representational Similarity for Training Neural Networks. Optimizing representations for high or low similarity during model training is a reoccurring theme across deep learning, e.g., in knowledge distillation [49] or fields that use contrastive representation learning [204, 230, 225, 191, 182].

The approaches to assessing similarity in this context are different from the similarity measures in this survey. First, differentiability and computational efficiency become important properties to enable gradient descent-based optimization. Second, and more importantly, in these processes it is often assumed that representations lie in the same representation space [194, 192, 199], and similarity is often evaluated on the instance level rather than on full representation matrices [213, 229, 193]. Alternatively, if the representations come from different models such as in multi-modal representation learning, their mapping into the joint space can be trained together with the rest of the system [229]. Hence, unless invariances of the similarity measure should be used to make optimization more flexible, invariances are not important. While these approaches are useful for training and aggregating representations at fixed layers of models, they do not generalize to post-training analysis of neural networks.

Distinction to *Functional Representations*. This survey covers representational and functional similarity measures. The terms *representational* and *functional* should not be confused with *functional representations*. In contrast to our work, which is about comparing neural networks, work on functional representations is about training neural networks to represent continuous functions that are only known via samples at discrete points [207, 218, 196].

F Orthogonal Procrustes and Dimensionality

To demonstrate how the similarity scores of a measure may be influenced by external factors such as dimensionality, we plot values of the Orthogonal Procrustes measure over varying dimension in Figure 5.

We compare two synthetic representation matrices: the first matrix is a random matrix with entries drawn from a standard normal distribution, and the second matrix is generated by multiplying the first matrix with an orthogonal matrix that was randomly drawn from the Haar distribution as implemented by `scipy`[4] [209], with added noise, that is again drawn from a normal distribution. These matrices have $N = 1000$ rows and varying dimension $D \in \{10, 50, 100, 200, 300, 1000, 2000, 3000\}$. This matrix generation process is repeated ten times for each value D, and we report the mean orthogonal Procrustes distance resulting from these matrix pairs. In addition, we create a baseline similarity score by permuting the rows of a copy of the original representation matrix, and comparing it to the original representation matrix, similar to the technique proposed by Kriegeskorte et al. [74]. We compute the baseline scores by shuffling the rows ten times for each representation pair, again reporting the mean.

The code to this experiment is available on GitHub[5].

[4] `https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.ortho_group.html`
[5] `https://github.com/mklabunde/survey_measures`

Table 6: *Models and Datasets that were considered in each test.* We separate models and datasets by domain (language, vision, or graphs). Checkmarks indicate that a model or dataset has been used in the corresponding test, otherwise cells are empty. For the *dimension subsample*, the *signal to noise matching*, and *cluster count* tests, no models have been used. Similarity measures were directly applied on the raw neuroimage data for the dimension subsample test, the other two used synthetic data.

		Models																					Datasets																		
		Language						Vision											Graphs				Language							Vision						Graphs			Other		
Test	Ref.	Transformer [226]	BERT [33]	ALBERT [201]	SmolLM2 [166]	GPT-2 [211]	XLM-R [176]	CNN/All-CNN-C [221]	AlexNet [198]	Inception [222]	VGG [217]	ResNet	MobileNet [188]	MnasNet [223]	RegNet [212]	EfficientNet [224]	ConvNeXt [202]	ViT [180]	GCN [195]	GraphSAGE [185]	GAT [227]	P-GNN [233]	MNLI [231]	SST-2 [219]	QNLI [228]	Penn TreeBank [205]	WikiText2 [208]	HANS [206]	WMT 2014 [171]	ImageNet [179]	CIFAR-10 [197]	CIFAR-100 [197]	SVHN [210]	XNLI [175]	UTKFace [235]	Cora [232]	Flickr [234]	OGBN-Arxiv [189]	Conceptual Captions [216]	Neuroimages [186]	Synthetic Data
Accuracy Correlation	[17]		✓																				✓	✓	✓			✓													
	[34]		✓									✓											✓	✓	✓						✓	✓	✓								
	[57]								✓			✓	✓																	✓											
	[70]		✓	✓	✓						✓	✓						✓	✓	✓	✓	✓	✓	✓						✓		✓				✓	✓	✓			
Disagreement Correlation	[8]										✓	✓																			✓	✓									
	[70]		✓	✓	✓						✓	✓						✓	✓	✓	✓	✓	✓	✓						✓		✓				✓	✓	✓			
JSD Correlation	[70]		✓	✓	✓						✓	✓						✓	✓	✓	✓	✓	✓	✓						✓		✓				✓	✓	✓			
Squared Error Correlation	[17]								✓	✓	✓	✓	✓	✓	✓	✓	✓																		✓						
Noise Addition	[101]																																								✓
Layer Matching	[24]							✓			✓	✓																		✓											
	[73]	✓						✓																					✓		✓										
	[116]		✓															✓								✓	✓				✓	✓									
	[125]							✓																							✓										
Dimension Subsample	[125]																																							✓	
Cluster Count	[8]																																								✓
Architecture Clustering	[17]								✓	✓	✓	✓	✓	✓	✓	✓	✓													✓											
Multilingual	[116]		✓				✓																											✓							
Image Caption	[116]		✓			✓											✓	✓																					✓		
Shortcut Affinity	[70]		✓	✓	✓						✓	✓						✓	✓	✓	✓	✓	✓	✓						✓		✓				✓	✓	✓			
Augmentation	[70]		✓	✓							✓	✓						✓	✓	✓	✓	✓	✓	✓						✓		✓				✓	✓	✓			
Label Randomization	[70]		✓	✓	✓						✓	✓						✓	✓	✓	✓	✓	✓	✓						✓		✓				✓	✓	✓			
Layer Monotonicity	[70]		✓	✓	✓						✓	✓						✓	✓	✓	✓	✓	✓	✓						✓		✓				✓	✓	✓			